



Lecture-3 (Relational Algebra)

Relation Schema and Relation Instance

Relation Schema

A **relation schema** is the *structure* or *blueprint* of a table in a database.

It tells:

- The **name** of the relation (table)
- The **attributes** (columns)
- The **domain** (type of data in each column)

✓ Example of Relation Schema

```
Student(StudentID, Name, Dept, Age)
```

Here:

- Relation name: **Student**
- Attributes: **StudentID, Name, Dept, Age**
- Domain example:
 - StudentID → integer
 - Name → string
 - Dept → string
 - Age → integer

This is just the *design*, no data inside yet.

Relation Instance

A **relation instance** is the *actual data* stored in the table **at a specific moment**.
It is a **set of rows (tuples)** that follow the schema.

✓ Example of Relation Instance

Student			

StudentID	Name	Dept	Age

101	Ali	CSE	20
102	Rahim	EEE	22
103	Tanvir	CSE	19

This is an **instance** of the Student relation.

Keys in a Database

Keys are used to **uniquely identify records** and maintain **relationships** between tables.

Below are the main types of keys with short, easy examples.

1 Primary Key

A primary key **uniquely identifies each row**.

No duplicates, no NULL values.

Example:

Student(StudentID, Name, Dept)

StudentID → Primary Key

Why?

Because every student has a unique ID.

2 Candidate Key

A candidate key is any attribute that **can** uniquely identify rows.

From these, we choose **one** as the primary key.

Example:

```
Employee(EmpID, Email, Phone)
```

Candidate Keys: EmpID, Email, Phone

Primary Key: EmpID (chosen)

3 Super Key

A super key is any set of attributes that uniquely identifies a row —
it may contain extra unnecessary attributes.

Example:

```
(StudentID), (StudentID, Name), (StudentID, Dept)
```

All are **super keys** because StudentID already identifies uniquely.

4 Alternate Key

The remaining candidate keys after choosing the primary key.

Example:

Primary Key = EmpID

Alternate Keys = Email, Phone

5 Foreign Key

A foreign key is used to **link two tables**.

It references a primary key in another table.

Example:

```
Department(DeptID, DeptName)
Student(StudentID, Name, DeptID) ← Foreign Key
```

`Student.DeptID` refers to `Department.DeptID`.

6 Composite Key

A key made from **two or more attributes**.

Example:

```
CourseEnrollment(StudentID, CourseID, Semester)

Composite Primary Key = (StudentID, CourseID)
```

Together they identify a unique enrollment.

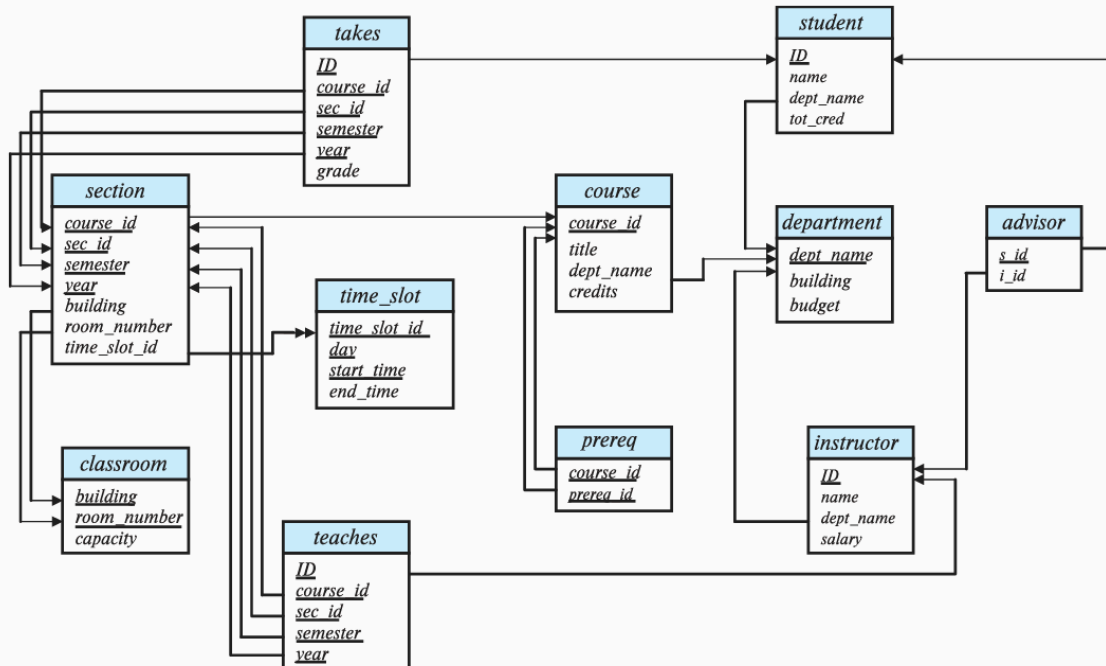
Summary Table

Key Type	Meaning	Small Example
Primary Key	Unique identifier	StudentID
Candidate Key	Can uniquely identify	Email, Phone
Super Key	Unique + extra attributes	(ID, Name)
Alternate Key	Candidate key not chosen	Email
Foreign Key	Links tables	DeptID
Composite Key	Multiple attributes	(StudentID, CourseID)

Schema Diagram for University Database



Schema Diagram for University Database



Database System Concepts - 7th Edition

2.9

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Relational Algebra

Relational algebra is a **theoretical language** used to **query** and **manipulate** data in a relational database.

It works on **relations (tables)** and produces **new relations (tables)** as results.

Think of it as the **mathematical foundation of SQL**.

1. Selection (σ)

Selection (σ) is a relational algebra operation used to:

- **Filter rows** of a table

- Based on a **condition**
- Only the rows that satisfy the condition appear in the result

Instructor Table

Instructor

```
-----
ID   | Name   | Dept  | Salary | Experience
-----
```

```
101  | Ali    | CSE   | 55000  | 5
102  | Rahim  | EEE   | 60000  | 7
103  | Sumi   | CSE   | 52000  | 3
104  | Tanvir | BBA   | 45000  | 2
105  | Jamil  | CSE   | 65000  | 10
```

1. Selection Operation (σ)

1 Example: Instructors from the CSE department

σ Dept = 'CSE' (Instructor)

Result:

```
-----
ID   | Name   | Dept  | Salary | Experience
-----
```

```
101  | Ali    | CSE   | 55000  | 5
103  | Sumi   | CSE   | 52000  | 3
105  | Jamil  | CSE   | 65000  | 10
```

2 Example: Instructors with salary greater than 55,000

σ Salary > 55000 (Instructor)

Result:

ID	Name	Dept	Salary	Experience
102	Rahim	EEE	60000	7
105	Jamil	CSE	65000	10

3 Example: CSE instructors with more than 4 years experience

σ (Dept = 'CSE' AND Experience > 4) (Instructor)

Result:

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
105	Jamil	CSE	65000	10

2. Projection (π)

Projection (π) is used to **select specific columns** from a table.

- It removes **unwanted attributes**
- It may also remove **duplicate rows**
- Output is a **new table** with only selected columns

Instructor Table (same as before)

Instructor

ID	Name	Dept	Salary	Experience

101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3
104	Tanvir	BBA	45000	2
105	Jamil	CSE	65000	10

Example: Select only Name and Salary

π (Name, Salary) (Instructor)

Result:

Name	Salary

Ali	55000
Rahim	60000
Sumi	52000
Tanvir	45000
Jamil	65000

Example: Select only Dept

π (Dept) (Instructor)

Result:

```
Dept
-----
CSE
EEE
CSE
BBA
CSE
```

After removing duplicates (relational algebra removes duplicates automatically):

```
Dept
-----
CSE
EEE
BBA
```

3. Union ($\cup$)

Union combines the rows of two relations (tables) and returns:

- All rows from **both tables**
- **No duplicate rows**
- Both tables must have the **same attributes** (same columns)

Union = A + B (without duplicates)

Requirements for Union

Two tables must be:

- **Union-compatible**
- Same number of columns

- Same data types
- Same order of attributes

Example Tables

Let's create two relations of **InstructorCSE** and **InstructorEEE**.

InstructorCSE

ID	Name	Dept
101	Ali	CSE
103	Sumi	CSE
105	Jamil	CSE

InstructorEEE

ID	Name	Dept
102	Rahim	EEE
106	Nafi	EEE

Example: Combine all instructors from CSE and EEE

`InstructorCSE ∪ InstructorEEE`

Result:

ID	Name	Dept
101	Ali	CSE
103	Sumi	CSE
105	Jamil	CSE

102		Rahim		EEE
106		Nafi		EEE

All rows from both tables, **no duplicates**.

2 Example: Union with overlapping rows

Let's modify InstructorEEE slightly:

InstructorEEE (With one duplicate row)

ID		Name		Dept

102		Rahim		EEE
103		Sumi		CSE

← Duplicate of InstructorCSE

Union:

InstructorCSE $\cup$ InstructorEEE

Result (duplicate removed automatically):

ID		Name		Dept

101		Ali		CSE
103		Sumi		CSE
105		Jamil		CSE
102		Rahim		EEE

4. Set Difference (-)

What is Set Difference?

Set Difference returns the rows that are in the **first relation** but **not in the second relation**.

- Denoted as: $R - S$
- Both tables must be **union-compatible** (same columns & types)
- Removes rows from the first table that are **also in the second table**

" $R - S$ = rows in R but not in S"

Example Tables

InstructorCSE

ID	Name	Dept
101	Ali	CSE
103	Sumi	CSE
105	Jamil	CSE

InstructorHighSalary

ID	Name	Dept
105	Jamil	CSE
101	Ali	CSE

Set Difference Operation

Example: Find CSE instructors not in high salary group

InstructorCSE – InstructorHighSalary

Result:

ID	Name	Dept
103	Sumi	CSE

✓ Only **Sumi** is in `InstructorCSE` but not in `InstructorHighSalary`.

Example: Difference with no common rows

Suppose:

InstructorEEE

ID	Name	Dept
102	Rahim	EEE
106	Nafi	EEE

`InstructorCSE` – `InstructorEEE`

Result:

ID	Name	Dept
101	Ali	CSE
103	Sumi	CSE
105	Jamil	CSE

✓ All rows remain because **no overlap** with `InstructorEEE`.

5. Cartesian Product (×)

✓ What is Cartesian Product?

- The Cartesian Product combines **every row of the first table** with **every row of the second table**.
- Denoted as: $R \times S$
- The result has **all combinations** of rows from both tables.

👉 In simple words:

"Pair each row of Table1 with each row of Table2."

Table1

```
ID | Name
-----
1  | Ali
2  | Babu
3  | Sumi
```

Table2

```
Dept | Salary
-----
CSE  | 50000
EEE  | 60000
BBA  | 45000
```

Cartesian Product

Table1 $\times$ Table2

Result:

```
ID | Name | Dept | Salary
-----
```

1		Ali		CSE		50000
1		Ali		EEE		60000
1		Ali		BBA		45000
2		Babu		CSE		50000
2		Babu		EEE		60000
2		Babu		BBA		45000
3		Sumi		CSE		50000
3		Sumi		EEE		60000
3		Sumi		BBA		45000

- ✓ Total rows = $3 \times 3 = 9$ rows
- ✓ Every row of Table1 is paired with every row of Table2.

6. Rename (ρ)

- **Rename (ρ)** is used to **give a new name** to:
 - A **relation (table)**
 - Or **attributes (columns)**
- Syntax:

```

ρ(NewRelationName, OldRelationName)
ρ(NewRelationName(NewAttribute1, NewAttribute2, ...), OldRelationName)1

```

- Useful when:
 - Joining tables with the **same attribute names**
 - Making queries **easier to read**

Example Table

Instructor Table

ID	Name	Dept	Salary
101	Ali	CSE	55000
102	Rahim	EEE	60000
103	Sumi	CSE	52000

Examples of Rename

1 Rename the table

```
ρ(Faculty, Instructor)
```

Result: Table is now called `Faculty` with same columns:

ID	Name	Dept	Salary
101	Ali	CSE	55000
102	Rahim	EEE	60000
103	Sumi	CSE	52000

2 Rename columns

```
ρ(Faculty(FacID, FacName, Department, Pay), Instructor)
```

Result: Columns renamed:

FacID	FacName	Department	Pay
101	Ali	CSE	55000
102	Rahim	EEE	60000
103	Sumi	CSE	52000

- ✓ Makes queries **clearer** and avoids **name conflicts**.

For Practice....

Selection (σ) with AND, OR, NOT

We use **AND**, **OR**, **NOT** to combine conditions in selection:

- **AND** → Both conditions must be true
- **OR** → Either condition is true
- **NOT** → Negates the condition

Instructor Table

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3
104	Tanvir	BBA	45000	2
105	Jamil	CSE	65000	10

3 Examples of AND, OR, NOT

1 AND Example

Select CSE instructors with salary > 55000

σ (Dept='CSE' AND Salary>55000) (Instructor)

Result:

ID	Name	Dept	Salary	Experience
----	------	------	--------	------------

```
105 | Jamil | CSE | 65000 | 10
```

2 OR Example

Select instructors from CSE **or** salary > 60000

```
σ (Dept='CSE' OR Salary>60000) (Instructor)
```

Result:

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
103	Sumi	CSE	52000	3
105	Jamil	CSE	65000	10
102	Rahim	EEE	60000	7

3 NOT Example

Select instructors **not in CSE**

```
σ NOT (Dept='CSE') (Instructor)
```

Result:

ID	Name	Dept	Salary	Experience
102	Rahim	EEE	60000	7
104	Tanvir	BBA	45000	2

Problem 1

Find all instructors from **BBA** or having **experience > 5 years**.

Solution:

```
σ (Dept='BBA' OR Experience>5) (Instructor)
```

Result:

102		Rahim		EEE		60000		7
104		Tanvir		BBA		45000		2
105		Jamil		CSE		65000		10

Problem 2

Find all instructors **not from EEE** and having **salary < 60000**.

Solution:

```
σ (NOT Dept='EEE' AND Salary<60000) (Instructor)
```

Result:

101		Ali		CSE		55000		5
103		Sumi		CSE		52000		3
104		Tanvir		BBA		45000		2

Problem 3

Select instructors from **CSE or BBA** but **not having experience < 5**.

Solution:

```
σ ((Dept='CSE' OR Dept='BBA') AND NOT (Experience<5)) (Instructor)
```

Result:

101		Ali		CSE		55000		5
105		Jamil		CSE		65000		10

Tables

Instructor

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3
104	Tanvir	BBA	45000	2
105	Jamil	CSE	65000	10

Department

Dept	Head
CSE	Dr. Karim
EEE	Dr. Rahman
BBA	Dr. Sultana

Question 1: Selection + Projection

Q: Find the **names and salaries** of instructors in **CSE** with **salary > 55000**.

Solution:

$$\pi (\text{Name, Salary}) (\sigma (\text{Dept} = \text{'CSE'} \text{ AND } \text{Salary} > 55000) (\text{Instructor}))$$

Result:

Name	Salary
------	--------

Jamil | 65000

Question 2: Union

Q: Find all instructors from **CSE or EEE**.

Solution:

$\sigma \text{ Dept} = \text{'CSE'} \text{ (Instructor)} \cup \sigma \text{ Dept} = \text{'EEE'} \text{ (Instructor)}$

Result:

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
103	Sumi	CSE	52000	3
105	Jamil	CSE	65000	10
102	Rahim	EEE	60000	7

Question 3: Set Difference

Q: Find all instructors **not in CSE**.

Solution:

$\text{Instructor} - \sigma \text{ Dept} = \text{'CSE'} \text{ (Instructor)}$

Result:

ID	Name	Dept	Salary	Experience
102	Rahim	EEE	60000	7
104	Tanvir	BBA	45000	2

Question 4: Cartesian Product

Q: Combine **Instructor** and **Department** to list **all possible combinations** of instructors and departments.

Solution:

```
Instructor × Department
```

Result: (Partial example)

ID	Name	Dept	Salary	Experience	Dept	Head
101	Ali	CSE	55000	5	CSE	Dr. Karim
101	Ali	CSE	55000	5	EEE	Dr. Rahman
101	Ali	CSE	55000	5	BBA	Dr. Sultan
102	Rahim	EEE	60000	7	CSE	Dr. Karim

Question 5: Rename

Q: Rename **Instructor** table to **Faculty** and **columns** as **FacID, FacName, Department, Pay, Exp**.

Solution:

```
ρ(Faculty(FacID, FacName, Department, Pay, Exp), Instructor)
```

Result: Table now appears as:

FacID	FacName	Department	Pay	Exp
101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3

104		Tanvir		BBA		45000		2
105		Jamil		CSE		65000		10

Question 6: Combined Operations

Q: Find the **names of CSE instructors** and their **department heads** by combining selection, projection, and join.

Solution:

1. Select CSE instructors:

```
 $\sigma$  Dept='CSE' (Instructor)
```

1. Join with Department table on Dept:

```
 $\sigma$  Dept='CSE' (Instructor)  $\bowtie$  Instructor.Dept = Department.Dept  
Department
```

1. Project Name and Head:

```
 $\pi$  (Name, Head) ( $\sigma$  Dept='CSE' (Instructor)  $\bowtie$  Instructor.Dept =  
Department.Dept Department)
```

Result:

Name		Head

Ali		Dr. Karim
Sumi		Dr. Karim
Jamil		Dr. Karim

Find Departments with No Instructor

We have:

Instructor Table

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3
104	Tanvir	BBA	45000	2
105	Jamil	CSE	65000	10

Department Table

Dept	Head
CSE	Dr. Karim
EEE	Dr. Rahman
BBA	Dr. Sultana
LAW	Dr. Ahmed
MED	Dr. Shirin

We want departments **that do not have any instructor assigned**.

Relational Algebra Solution

We can use **Set Difference (-)**:

1. Get all departments from Instructor table:

$$\pi (\text{Dept}) (\text{Instructor})$$

Result:

CSE
EEE

BBA

1. Get all departments from Department table:

π (Dept) (Department)

Result:

CSE
EEE
BBA
LAW
MED

1. Subtract:

π (Dept) (Department) – π (Dept) (Instructor)

Result

LAW
MED

✓ These are departments with **no instructor**.

Practice Questions – Instructor & Department Tables

Tables

Instructor

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
102	Rahim	EEE	60000	7
103	Sumi	CSE	52000	3
104	Tanvir	BBA	45000	2
105	Jamil	CSE	65000	10

Department

Dept	Head
CSE	Dr. Karim
EEE	Dr. Rahman
BBA	Dr. Sultana
LAW	Dr. Ahmed
MED	Dr. Shirin

Questions

Q1: Find departments with instructors having salary > 60000.

Solution:

$$\pi (\text{Dept}) (\sigma \text{ Salary} > 60000 (\text{Instructor}))$$

Result:

CSE

Q2: Find departments that have more than 1 instructor.

Solution (using selection + projection):

```
 $\pi$  (Dept) ( $\sigma$  Dept='CSE' OR Dept='EEE' OR Dept='BBA' (Instructor))
```

Result:

CSE

(Since CSE has 3 instructors, others have 1)

Q3: List instructors and their department heads.

Solution:

```
Instructor  $\bowtie$  Instructor.Dept = Department.Dept Department
```

Result:

ID	Name	Dept	Salary	Experience	Head
101	Ali	CSE	55000	5	Dr. Karim
102	Rahim	EEE	60000	7	Dr. Rahman
103	Sumi	CSE	52000	3	Dr. Karim
104	Tanvir	BBA	45000	2	Dr. Sultana
105	Jamil	CSE	65000	10	Dr. Karim

Q4: Find departments with no instructor.

Solution:

```
 $\pi$  (Dept) (Department) -  $\pi$  (Dept) (Instructor)
```

Result:

LAW

MED

Q5: Find CSE instructors with salary > 50000.

Solution:

```
 $\sigma$  (Dept='CSE' AND Salary>50000) (Instructor)
```

Result:

ID	Name	Dept	Salary	Experience
101	Ali	CSE	55000	5
105	Jamil	CSE	65000	10

Q6: Find names of instructors and department heads for instructors with experience < 5.

Solution:

```
 $\pi$  (Name, Head) ( $\sigma$  Experience<5 (Instructor)  $\bowtie$  Instructor.Dept  
= Department.Dept Department)
```

Result:

Name	Head
Sumi	Dr. Karim
Tanvir	Dr. Sultana